Nptel Online Certification Course: Computer Vision
Week 9: Range image processing
Type of questions: MCQ/MSQ
Number of questions: 10 **Total marks: 10 X 2 = 20**

QUESTION 1: **Type: MCQ**

In binary (Gray-code) structured-light encoding, n patterns are projected. How many distinct stripes can be uniquely encoded?

a) n

b) $2n$

c) 2^n

d) n^2

Correct Answer: c
Solution: Each stripe receives a unique n-bit binary codeword from the n projected black-and-white patterns. With n bits there are exactly 2^n distinct codewords, so 2^n stripes can be encoded.

QUESTION 2: **Type: MCQ**

In a 3-pattern binary structured-light system, what is the 3-bit *Gray code* assigned to stripe number 5 (0-indexed)?

a) 101

b) 111

c) 110

d) 100

Correct Answer: b
Solution: The Gray code of an integer n is obtained by $g = n \oplus (n \gg 1)$ (XOR with its own right-shift by 1):

$$5_{10} = 101_2, \quad g = 101 \oplus 010 = 111_2.$$

QUESTION 3: **Type: MCQ**

Which of the following statements about the Microsoft Kinect (first generation) sensor is/are correct?

a) It projects an infrared laser speckle pattern onto the scene.

b) Points at greater depth appear more defocused / blurry in the IR image.

c) It operates on the principle of structured light triangulation.

d) All of the above.

Correct Answer: d

Solution: The first-generation Kinect uses an infrared (IR) laser diode that projects a pseudo-random speckle pattern. A separate IR camera observes how the speckle deforms with depth (structured-light triangulation). Because the IR camera has a fixed focus, objects at depths further from the focal plane produce blurrier speckle spots. All three statements are therefore true.

QUESTION 4: **Type:MCQ**

Consider the surface $z = xy$ parametrised as $\mathbf{X}(u, v) = (u,\ v,\ uv)$. Which of the following is the first fundamental form?

a) $(1 + v^2)\,du^2 + 2uv\,du\,dv + (1 + u^2)\,dv^2$

b) $(1 + v)\,du^2 + uv\,du\,dv + (1 + u)\,dv^2$

c) $(1 + v^3)\,du^2 + 2uv\,du\,dv + (1 + u^3)\,dv^2$

d) $(1 + v^2)\,du^2 + uv\,du\,dv + (1 + u^2)\,dv^2$

Correct Answer: a

Detailed Solution: Compute the partial derivatives:

$$\mathbf{X}_u = (1,\ 0,\ v), \qquad \mathbf{X}_v = (0,\ 1,\ u).$$

First fundamental form coefficients:

$$E = \mathbf{X}_u \cdot \mathbf{X}_u = 1 + v^2, \quad F = \mathbf{X}_u \cdot \mathbf{X}_v = uv, \quad G = \mathbf{X}_v \cdot \mathbf{X}_v = 1 + u^2.$$

The first fundamental form is:

$$ds^2 = E\,du^2 + 2F\,du\,dv + G\,dv^2 = (1 + v^2)\,du^2 + 2uv\,du\,dv + (1 + u^2)\,dv^2.$$

QUESTION 5: **Type: MCQ**

For the surface $\mathbf{X}(u, v) = (u,\ v^2,\ uv)$, compute the first fundamental form coefficients E, F, G at the point $(u, v) = (1, 2)$.

a) $E = 5,\ F = 2,\ G = 17$

b) $E = 5,\ F = 4,\ G = 9$

c) $E = 2,\ F = 4,\ G = 17$

d) $E = 5,\ F = 2,\ G = 5$

Correct Answer: a
Detailed Solution:

$$\mathbf{X}_u = (1,\ 0,\ v), \qquad \mathbf{X}_v = (0,\ 2v,\ u).$$

At $(u, v) = (1, 2)$: $\mathbf{X}_u = (1, 0, 2)$, $\mathbf{X}_v = (0, 4, 1)$.

$$\begin{aligned} E &= \mathbf{X}_u \cdot \mathbf{X}_u = 1 + 0 + 4 = 5, \\ F &= \mathbf{X}_u \cdot \mathbf{X}_v = 0 + 0 + 2 = 2, \\ G &= \mathbf{X}_v \cdot \mathbf{X}_v = 0 + 16 + 1 = 17. \end{aligned}$$

QUESTION 6: **Type: MCQ**

Consider the 2D parametric curve $x(t) = t^2 + t,\ y(t) = 2t - 1$. Compute the tangent vector $(x'(t),\ y'(t))$ at $t = 1$.

a) $(2,\ 2)$

b) $(3,\ 2)$

c) $(2,\ 4)$

d) $(1,\ 2)$

Correct Answer: b
Detailed Solution:
Differentiate each component:

$$x'(t) = 2t + 1, \qquad y'(t) = 2.$$

At $t = 1$:

$$x'(1) = 2(1) + 1 = 3, \qquad y'(1) = 2.$$

Tangent vector = $(3,\ 2)$.

QUESTION 7: **Type: MCQ**

Compute the *direction* of the surface normal (i.e. $\mathbf{S}_u \times \mathbf{S}_v$ before normalisation) at $(u, v) = (1, 0)$ for the surface $\mathbf{S}(u, v) = (3u + v,\ u^2,\ 2v)$.

a) $(4,\ -6,\ -2)$

b) $(4,\ 6,\ -2)$

c) $(-4,\ -6,\ 2)$

d) $(4,\ -6,\ 2)$

Correct Answer: a
Detailed Solution:

$$\mathbf{S}_u = (3,\ 2u,\ 0) \xrightarrow{u=1} (3,\ 2,\ 0), \qquad \mathbf{S}_v = (1,\ 0,\ 2).$$

$$\mathbf{S}_u \times \mathbf{S}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 3 & 2 & 0 \\ 1 & 0 & 2 \end{vmatrix} = \mathbf{i}(4 - 0) - \mathbf{j}(6 - 0) + \mathbf{k}(0 - 2) = (4,\ -6,\ -2).$$

QUESTION 8: **Type: MCQ**

Given $E = 4,\ F = 1,\ G = 3,\ e = 5,\ f = 1,\ g = 2$, compute the mean curvature H (upto 4 decimal places).

a) $H = 0.8182$

b) $H = 0.9545$

c) $H = 1.0455$

d) $H = 0.7273$

Correct Answer: b
Detailed Solution:

$$H = \frac{Eg + Ge - 2Ff}{2(EG - F^2)} = \frac{(4)(2) + (3)(5) - 2(1)(1)}{2[(4)(3) - (1)^2]} = \frac{8 + 15 - 2}{2 \times 11} = \frac{21}{22} \approx \boxed{0.9545}.$$

QUESTION 9: **Type: MCQ**

A surface patch is measured at a point P and found to have principal curvatures $\kappa_1 = +3.2$ and $\kappa_2 = -1.7$. Which of the following correctly classifies the surface at P and gives the sign of its Gaussian curvature K?

a) Pit surface; $K > 0$

b) Saddle surface; $K < 0$

c) Ridge surface; $K = 0$

d) Flat surface; $K = 0$

Correct Answer: b

Detailed Solution: the Gaussian curvature is the product of the two principal curvatures:

$$K = \kappa_1 \cdot \kappa_2.$$

Since κ_1 and κ_2 have *opposite signs*,

$$K = (+3.2)(-1.7) = -5.44 < 0.$$

QUESTION 10: **Type: MCQ**

Which of the following statements about edge detection in range images is *correct*?

a) Step edges are detected from zero crossings of Gaussian curvature.

b) Roof edges are detected from zero crossings of Gaussian curvature.

c) Roof edge points are detected by extrema of the dominant principal curvature along its corresponding direction.

d) Step edge points are detected by extrema of the dominant principal curvature along its corresponding direction.

Correct Answer: c

Detailed Solution: In range image edge detection, step edges arise at depth discontinuities (jumps in the range value) and are detected by *zero crossings of Gaussian curvature* (not by extrema). Roof edges arise where two surfaces meet at a dihedral angle and are detected by *extrema of the dominant principal curvature* along the direction corresponding to that curvature.
